
Introduction to Web Tech
Objectives
Explain fundamental concepts of modern web technologies
Differentiate frontend & backend technologies
Describe RESTful APIs and HTTP methods
Frontend Technologies
Core: HTML, CSS, JavaScript
Frameworks: Vue.js, React.js, Angular
Features: Component-based UI, State Management, API Communication, Responsive Design, PWAs
RESTful APIs
GET, POST, PUT, DELETE Methods
CRUD Operations
Web Application Security
SQL Injection, XSS, CSRF
Prevention Techniques
Cloud Technologies
Providers: AWS, GCP, Azure, Vercel, DigitalOcean
Deployment: Containerization, Serverless, CI/CD
DevOps Practices
Tools: Docker, Kubernetes, Jenkins
Overview: Key Components
Frontend (Client-Side)
Backend (Server-Side)
Database (Storage)
JSON (Data Exchange)
RESTful APIs
Security
JWT (JSON Web Token)
OAuth2
Cloud Computing
CI/CD
DevOps
Backend Technologies
Frameworks: PHP Slim, Node.js (Express.js), Django, Spring Boot, ASP.NET
GraphQL
Functionalities: Routing, Authentication, Business Logic, Logging
Database Technologies
SQL Databases: MySQL, PostgreSQL, MS SQL Server, SQLite
NoSQL Databases: MongoDB, Firebase Firestore
Authentication (JWT & OAuth2)
JWT Workflow
OAuth2 Workflow
CI/CD
Automation of testing, building, and deployment